

Interference Scaling, Resource Isolation, and Topology Sensitivity in 5G NR Sidelink UAV Swarms

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Abstract—Direct NR sidelink can support infrastructure-independent UAV-swarm coordination, but its interference regime depends jointly on density, communication topology, resource reuse, and spatial selectivity. We present a reproducible system-level study that combines a published 3.5 GHz air-to-air path-loss fit, 3GPP-derived MCS/TBS/LDPC mechanics under an explicit simplified sidelink resource-grid profile, and generic NR link-level EESM BLER curves processed from the official 5G-LENA v5.0 release. For swarm size N , $\lfloor N/2 \rfloor$ one-way disjoint links are active per snapshot. A 100-seed campaign evaluates $N \in \{5, 10, 20, 30, 50, 75, 100\}$, partitions a 133-PRB 50 MHz/30 kHz profile over $R \in \{1, 2, 4, 8\}$ abstract orthogonal resources, and tests $G \in \{0, 3, 6, 9\}$ dB desired/interferer relative advantage. Under the full-load sequential-disjoint baseline, $N = 100$ yields -23.03 dB mean SINR, 0.0128 first-transmission success probability, and 0.359 Mbps expected PHY goodput; $R = 8, G = 6$ raises the latter metrics to 0.5292 and 2.229 Mbps. A separate 100-seed robustness campaign shows that topology is a first-order determinant in this model: greedy short-link pairing at $N = 100, R = 1, G = 0$ reduces mean desired-link length from 517 m to 84 m and raises success probability to 0.4766. The results therefore characterize topology-dependent, mean-based policy operating regions rather than a universal swarm-capacity limit.

Index Terms—5G NR sidelink, UAV swarm, interference, resource allocation, topology, link adaptation, reproducibility.

I. INTRODUCTION

Reliable wireless links are a core enabling technology for cooperative UAV systems, whose mobility and altitude create propagation, interference, and networking conditions that differ substantially from conventional terrestrial deployments [1], [2]. NR sidelink provides standardized direct-communication and resource-allocation mechanisms [3]. This makes sidelink a relevant candidate for intra-swarm communication, as investigated in UAV-swarm scheduling and suitability studies [4], [5].

Aerial links frequently exhibit strong line-of-sight components, which can strengthen both desired and interfering signals [1], [2]. Analytical multi-UAV work shows that node geometry and co-channel interference materially affect SINR and outage [6]. Related studies jointly optimize trajectory, frequency, and routing [7], adapt MCS for U2U sidelink [8], or address synchronization and Doppler-aware sidelink access [9]. We therefore examine how desired-link topology interacts with the bandwidth cost of resource isolation when reliability is

evaluated through NR-aware transport-block mechanics rather than a single SINR threshold.

This paper asks: *How do swarm density and communication topology change the evaluated NR-sidelink interference regime, and how far can resource separation and spatial desired/interferer advantage move the resulting reliability/goodput operating region?* The contributions are threefold. First, we combine measurement-derived A2A path loss with 3GPP-derived MCS/TBS/LDPC mechanics under an explicit study resource grid and sourced 5G-LENA link-level BLER curves. Second, we quantify the trade-off between co-channel interference reduction and reduced per-link PRBs when the 133-PRB profile is partitioned into 1/2/4/8 resources. Third, matched-seed controls vary pairing, allocator, link adaptation, and noise-bandwidth accounting to identify which conclusions are robust and which are model-dependent. In particular, the severe high-density degradation of the long-link baseline is not topology invariant.

II. RELATED WORK AND RESEARCH GAP

UAV-communications surveys identify line-of-sight propagation, three-dimensional mobility, interference, and topology as central design variables [1], [2]; the NR-V2X literature documents sidelink physical-layer and autonomous resource-selection mechanisms [3]. UAV-specific work uses sidelink for communication-aware scheduling and multi-hop forwarding [4], evaluates NR sidelink for internal swarm communication [5], and studies dynamic U2U MCS control in ns-3 [8]. Complementary work derives multi-UAV interference/outage models [6], jointly optimizes trajectory, frequency, and routing [7], or targets synchronization and Doppler-aware sidelink access [9].

Our study does not claim a new standards-complete sidelink protocol. Instead, it isolates swarm size, desired-link topology, frequency-resource partitioning, a geometry-based reuse allocator, and an experimental spatial desired/interferer advantage while retaining NR-derived TBS/LDPC mechanics and sourced link-level BLER behavior. The key distinction is the matched-seed robustness layer: pairing, random versus geometry-aware allocation, adaptive versus fixed MCS, and noise-bandwidth accounting are varied explicitly rather than absorbed into a single optimized policy.

III. SYSTEM MODEL AND REPRODUCIBILITY

A. Propagation and link abstraction

The large-scale A2A channel uses the 3.5 GHz measurement-derived fit of Erdemir *et al.* [10],

$$PL(d) = 34.650 + 21.66 \log_{10}(d/1 \text{ m}). \quad (1)$$

Simulated RF values are model-derived, not field measurements. The study profile uses 50 MHz, 30 kHz SCS, 133 PRBs, and 0.5 ms slots. The 133-PRB bandwidth configuration is consistent with the NR UE table [11], while the 0.5-ms slot duration follows the 30-kHz NR numerology [12]. MCS values, TBS quantization, and LDPC segmentation follow NR procedures [13], [14], but the resource grid is an explicit study profile rather than a complete PSSCH implementation: each slot reserves two control/PSCCH-like symbols and one guard symbol, uses 11 PSSCH symbols, and removes 24 DM-RS RE/PRB. PSFCH, SL-PRS, and detailed first/second-stage SCI RE accounting are not modeled explicitly.

For each candidate MCS, the computed TBS determines base graph and nominal code-block size. The requested code-block size is mapped to the nearest available 5G-LENA curve with the same MCS/base graph; BLER is linearly interpolated in SINR and clamped to the sourced curve endpoints. Transport-block BLER is derived from code-block BLER under an independent-code-block approximation. These are generic NR EESM link-level simulation tables from 5G-LENA [15], [16], not measured UAV or PSSCH-specific BLER data. Link adaptation selects the available MCS maximizing expected first-transmission delivered bits per slot; a fixed-MCS-4 control tests this optimistic instantaneous-SINR assumption.

B. Topology and interference controls

For swarm size N , the simulator forms $\lfloor N/2 \rfloor$ one-way disjoint transmitter–receiver links; all formed links transmit simultaneously in each static snapshot and one UAV is unpaired when N is odd. The baseline pairs node indices $(0, 1), (2, 3), \dots$, so desired-link distance is independent of proximity. The robustness topology instead globally sorts pairwise distances and greedily selects the shortest non-overlapping pairs; we call this *greedy short-link pairing*, not a standardized association rule.

The 133 PRBs are partitioned as evenly as possible across $R \in \{1, 2, 4, 8\}$ abstract frequency resources. Different resources are assumed perfectly orthogonal, i.e., no adjacent-channel/inter-carrier leakage is modeled. For $R > 1$, the baseline reuse allocator forms the symmetric geometric conflict weight

$$w_{ij} = d^{-2}(\text{Tx}_i, \text{Rx}_j) + d^{-2}(\text{Tx}_j, \text{Rx}_i), \quad (2)$$

orders links by total conflict weight, and greedily assigns each link to the resource with minimum accumulated same-resource weight. This is a research abstraction, not NR Mode-1/Mode-2 sensing or scheduling [3]. The inverse-square conflict proxy is intentionally heuristic and is not identical to the fitted A2A path-loss exponent 2.166 used by the propagation model. Seeded

TABLE I
BASELINE STUDY CONFIGURATION.

Parameter	Value
Area / altitude	1000 × 1000 m / 100 m
Carrier / bandwidth / SCS	3.5 GHz / 50 MHz / 30 kHz
Tx power / noise figure	30 dBm / 7 dB
Grid profile	133 PRBs; 2 control + 1 guard; 11 PSSCH symbols
Active links per snapshot	$\lfloor N/2 \rfloor$, one-way, full load
Swarm sizes N	5, 10, 20, 30, 50, 75, 100
R / G	1, 2, 4, 8 / 0, 3, 6, 9 dB
Monte Carlo	100 matched deterministic seeds

random assignment is the no-coordination control. Directional relative advantage $G \in \{0, 3, 6, 9\}$ dB is imposed as $+G/2$ dB desired gain and $-G/2$ dB co-channel interference gain; it is a sensitivity variable, not a beamforming model.

C. Metrics, statistics, and evidence

First-TX success probability is $1 - \text{TB- BLER}$. Expected PHY goodput is $\text{TBS}(1 - \text{TB- BLER})/0.5$ ms under one scheduled TB per active link per slot; it excludes queues, higher-layer overhead, retransmission timing, and scheduler idling. Primary treatment comparisons reuse identical seeds/geometries and report paired differences, Cohen’s d_z , and paired percentile-bootstrap 95% intervals computed from 10,000 resamples with a fixed RNG seed for reproducibility.

The operating envelope is an explicit engineering-policy summary: a scenario is *mean-policy-qualified* when its scenario-mean first-TX success probability is at least 0.10 and mean expected PHY goodput is at least 1 Mbps. Thresholding uses mean point estimates, not lower confidence bounds, so the reported largest evaluated N is not a capacity guarantee.

Code, deterministic seed definitions, compact publication extracts, and workflow provenance are versioned in the public repository,¹ while full per-seed evidence is archived by the GitHub Actions artifact recorded in the repository. The audited publication run used commit f224ba9e5f95 and stores the manuscript snapshot, generated tables, and artifact digest together.

IV. RESULTS

A. Baseline interference and resource partitioning

Under sequential-disjoint pairing with one shared resource and $G = 0$, mean SINR decreases from -0.49 dB at $N = 5$ to -23.03 dB at $N = 100$. At $N = 100$ (50 simultaneous directed links), first-TX success probability is 0.0128 and expected PHY goodput is 0.359 Mbps. Using the explicit success-probability threshold 0.5, the mean per-seed conditional share among policy-failed links is 72.4% aggregate-interference limited and 27.6% dominant-interferer limited. This is a heuristic diagnostic, not a pooled-link fraction or a causal channel classification.

Table II shows that frequency-resource separation can improve reliability despite reducing the PRBs available to each link. Goodput is non-monotonic: $R = 4$ has higher mean SINR than $R = 2$ but lower goodput. This reflects the joint

¹<https://github.com/panagiotagrosdouli/uav-sidelink-swarm>

TABLE II
SELECTED $N = 100$ BASELINE OPERATING POINTS.

(R, G)	SINR (dB)	Success prob.	Goodput (Mbps)
(1, 0)	-23.03	0.0128	0.359
(2, 0)	-16.20	0.0749	0.668
(4, 0)	-10.48	0.0775	0.572
(8, 0)	-5.47	0.2049	1.042
(8, 6)	0.53	0.5292	2.229
(8, 9)	3.53	0.8050	3.493

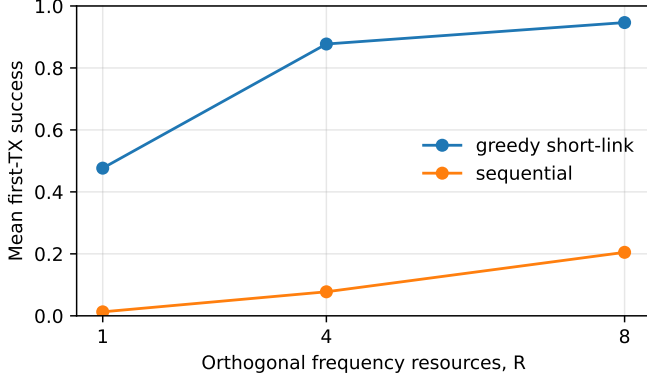


Fig. 1. Pairing sensitivity at $N = 100, G = 0$ with adaptive MCS and conflict-graph allocation.

effect of lower co-channel interference, changed conflict-graph assignments, and smaller per-link TBS; more resource groups do not guarantee more goodput. With $R = 8, G = 6$, first-TX success probability reaches 0.5292 and expected goodput 2.229 Mbps. Relative to $R = 1, G = 0$ at $N = 100$, the matched-seed differences are 0.5164 in success probability (paired-bootstrap 95% CI [0.5060, 0.5266], $d_z = 9.61$) and 1.870 Mbps in goodput (CI [1.721, 2.006], $d_z = 2.59$).

B. Topology is a first-order determinant in the model

At $N = 100, R = 1, G = 0$, sequential pairing produces a 517.20 m mean desired-link distance, whereas greedy short-link pairing reduces it to 84.05 m. Mean SINR changes from -23.03 to -1.75 dB, first-TX success probability from 0.0128 to 0.4766, and goodput from 0.359 to 15.116 Mbps. Fig. 1 shows that the difference persists as R increases. Thus, the severe degradation of the sequential baseline is not a topology-independent density law.

The same effect appears across swarm size. With $R = 1, G = 0$, sequential pairing keeps mean desired-link distance near 515–525 m from $N = 20$ to 100, while success probability falls from 0.0531 to 0.0128. Greedy short-link pairing reduces mean desired-link distance from 195.3 m at $N = 20$ to 84.0 m at $N = 100$; success probability changes from 0.5834 to 0.4766 despite the increasing number of simultaneous links (Fig. 2). The experiment therefore separates two competing geometric effects: more active transmitters increase interference, while denser local pairing strengthens the desired link.

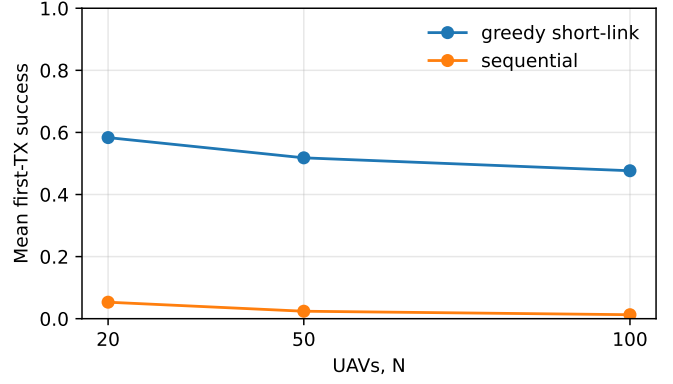


Fig. 2. Topology-conditioned scaling for $R = 1, G = 0$.

TABLE III
ROBUSTNESS CHECKS AT $N = 100$.

Dimension	Control / treatment	Main observation
Pairing, $R = 1, G = 0$	sequential $\rightarrow$ short-link	success 0.0128 $\rightarrow$ 0.4766
Allocator, $R = 8, G = 0$	random $\rightarrow$ conflict graph	goodput 0.815 $\rightarrow$ 1.042 Mbps
Link adaptation, $R = 8, G = 6$	fixed MCS-4 $\rightarrow$ adaptive	goodput 0.722 $\rightarrow$ 2.229 Mbps
Noise bandwidth	nominal share $\rightarrow$ exact PRB	max $ \Delta \text{SINR} = 0.113$ dB

C. Allocator and link-adaptation controls

Resource coordination adds benefit beyond increasing R . At $N = 100, R = 8, G = 0$, sequential pairing gives success/goodput 0.2049/1.042 Mbps with the conflict graph versus 0.1463/0.815 Mbps with random allocation. Under greedy short-link pairing, the corresponding values are 0.9465/10.836 and 0.8824/8.357 Mbps. Link adaptation is also material: at $N = 100, R = 8, G = 6$ under sequential pairing, adaptive MCS gives 0.5292/2.229 Mbps versus 0.3309/0.722 Mbps for fixed MCS-4. The adaptive result is therefore an optimistic system-level control case, not normative 3GPP AMC. By contrast, changing from nominal-channel PRB-share noise bandwidth to exact occupied PRB bandwidth changes SINR by at most 0.113 dB over the overlapping robustness grid, with negligible success/goodput change.

D. Mean-based policy operating envelope

For the explicit policy requiring scenario-mean first-TX success probability ≥ 0.10 and mean expected goodput ≥ 1 Mbps, Fig. 3 reports the largest *evaluated* N meeting both point-estimate thresholds for each (R, G) . The sequential baseline reaches $N = 10$; several resource/directionality combinations reach the evaluated $N = 100$ boundary. These points are policy-qualified operating points, not confidence-bound guarantees or universal capacity limits.

V. DISCUSSION AND LIMITATIONS

The results show that topology and resource reuse cannot be separated when interpreting dense aerial sidelink. Prior

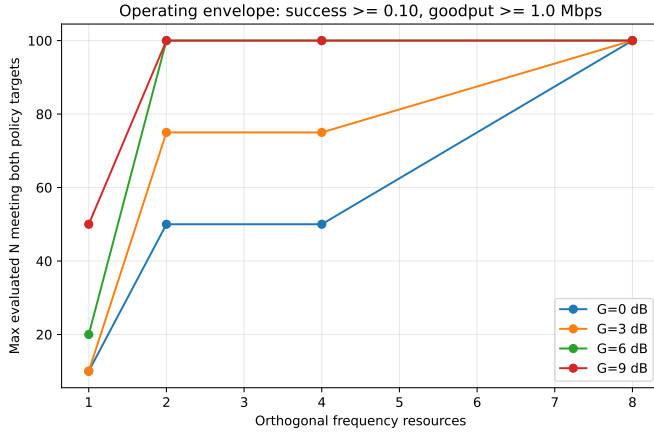


Fig. 3. Mean-based operating envelope for the sequential baseline under explicit policy thresholds.

multi-UAV work likewise couples interference with geometry, routing, and frequency allocation [6], [7]; recent U2U sidelink work also demonstrates that MCS adaptation materially affects throughput/reliability trade-offs [8]. Our controls show the same dependence: resource partitioning helps the long-link baseline, but shortening desired links changes the shared-resource regime more strongly.

Orthogonalization is not free. The $R = 2$ versus $R = 4$ result reflects the joint effect of conflict-graph placement, lower co-channel interference, and reduced TBS as PRBs are split. Accordingly, the operating envelope is a map for one explicit topology/load/model combination rather than a scalar maximum swarm size.

A. Threats to validity

Propagation domain: the fitted A2A law comes from a 3.5 GHz, 50 MHz, 30 dBm campaign with both drones at 100 m, an initial separation of 85 m, and a roughly 1 km A2A flight path in a mostly hilly/forested, rural-like campus environment [10]. Our carrier, bandwidth, altitude, and transmit power match those conditions, but the synthetic square permits link lengths up to 1.41 km, whereas the measurement campaign reports a 1-km transmitter trajectory starting from 85 m separation. The longest simulated links may therefore extend beyond the measurement campaign’s sampled separation geometry. Shadowing correlation, fast fading, blockage, Doppler, antenna-body effects, and temporal channel memory are omitted.

PHY/MAC abstraction: the study uses 3GPP-derived MCS/TBS/LDPC rules but an explicit simplified resource grid. PSFCH, SL-PRS, detailed SCI resource accounting, feedback timing, retransmissions, power control, and standards-complete sensing/resource selection are outside the paper model. Different abstract frequency resources are perfectly orthogonal. The 5G-LENA curves are generic NR EESM link-level tables [15], [16]; requested code-block size is mapped to the nearest sourced same-MCS/base-graph curve, SINR is linearly interpolated and endpoint-clamped, and TB-BLER

uses an independent-code-block approximation. These choices bound the absolute goodput interpretation.

Traffic/topology abstraction: each snapshot contains $\lfloor N/2 \rfloor$ simultaneously active one-way links; it does not model bidirectional application traffic, queues, mobility during a packet exchange, or mission-constrained partner selection. Greedy short-link pairing is a sensitivity control, not a proposed association protocol. Likewise, G is a desired/interferer power-ratio sensitivity, not an antenna-array or beam-tracking model.

Statistics and policy: matched seeds reduce geometry variance in treatment comparisons, but the operating-envelope thresholds are engineering choices applied to scenario means. They do not require the lower confidence bound to satisfy the target and therefore should not be interpreted as reliability guarantees.

B. Design implications

Within these boundaries, four conclusions are supported by the evaluated model: (i) desired-link topology must be specified before interpreting density scaling; (ii) more frequency groups do not necessarily improve goodput because interference reduction competes with PRB/TBS loss; (iii) geometry-aware reuse outperforms random allocation in the tested cases; and (iv) link adaptation and spatial selectivity complement rather than replace topology/resource design. Future work should repeat the matched-seed methodology with standards-complete sidelink sensing/scheduling, richer channel dynamics, realistic bidirectional traffic, and multi-UAV RF validation.

VI. CONCLUSION

Under the evaluated full-load sequential-disjoint topology, increasing N increases the number of simultaneous directed links and drives the shared-resource case into a severe interference-limited regime. Resource partitioning and the experimental desired/interferer advantage move the mean-based operating region substantially. The robustness campaign also shows why this result must be topology-conditioned: at $N = 100$, greedy short-link pairing raises shared-resource first-TX success probability from 0.0128 to 0.4766 by shortening desired links. Geometry-aware reuse and adaptive MCS provide additional gains, while the tested noise-bandwidth convention has negligible effect. The supported design message is therefore conditional but actionable: topology, reuse, spatial selectivity, and link adaptation should be evaluated jointly rather than inferring swarm capacity from N alone.

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REFERENCES

- [1] Y. Zeng, Q. Wu, and R. Zhang, "Accessing From the Sky: A Tutorial on UAV Communications for 5G and Beyond," *Proc. IEEE*, vol. 107, no. 12, pp. 2327–2375, Dec. 2019, doi: 10.1109/JPROC.2019.2952892.
- [2] M. Mozaffari, W. Saad, M. Bennis, Y.-H. Nam, and M. Debbah, "A Tutorial on UAVs for Wireless Networks: Applications, Challenges, and Open Problems," *IEEE Commun. Surveys Tuts.*, vol. 21, no. 3, pp. 2334–2360, 2019, doi: 10.1109/COMST.2019.2902862.
- [3] M. H. Castañeda García, A. Molina-Galan, M. Boban, J. Gozalvez, B. Coll-Perales, T. Şahin, and A. Kousaridas, "A Tutorial on 5G NR V2X Communications," *IEEE Commun. Surveys Tuts.*, vol. 23, no. 3, pp. 1972–2026, 2021, doi: 10.1109/COMST.2021.3057017.
- [4] D. Mishra, A. Trotta, E. Traversi, M. Di Felice, and E. Natalizio, "Cooperative Cellular UAV-to-Everything (C-U2X) communication based on 5G sidelink for UAV swarms," *Computer Communications*, vol. 192, pp. 173–184, 2022, doi: 10.1016/j.comcom.2022.06.001.
- [5] M. Varonen, "Using NR Sidelink for UAV Swarm Internal Communication," M.S. thesis, School of Electrical Engineering, Aalto Univ., Espoo, Finland, 2024, URN:NBN:fi:aalto-202403172777.
- [6] W. J. Lau, J. M.-Y. Lim, C. Y. Chong, N. S. Ho, and T. W. M. Ooi, "General outage probability model for UAV-to-UAV links in multi-UAV networks," *Computer Networks*, vol. 229, Art. no. 109752, 2023, doi: 10.1016/j.comnet.2023.109752.
- [7] M. M. Alam and S. Moh, "Joint trajectory control, frequency allocation, and routing for UAV swarm networks: A multi-agent deep reinforcement learning approach," *IEEE Trans. Mobile Comput.*, vol. 23, no. 12, pp. 11989–12005, 2024, doi: 10.1109/TMC.2024.3403890.
- [8] J. Yu and H. Jiang, "ns3-uavlink: AI-Driven Dynamic MCS Scheduling for U2U Sidelink Communication," in *Proc. IEEE VTC2025-Fall*, pp. 1–5, 2025, doi: 10.1109/VTC2025-Fall65116.2025.11310275.
- [9] H. Zhang, H.-M. Chen, Q.-J. Wei, Z.-W. Wang, and Y.-H. Sun, "Optimized synchronization design for UAV swarm network based on sidelink," *Drones*, vol. 10, no. 4, Art. no. 304, 2026, doi: 10.3390/drones10040304.
- [10] U. Erdemir, B. Kaplan, I. Hokelek, A. Gorcin, and H. A. Cirpan, "Measurement-based channel characterization for A2A and A2G wireless drone communication systems," in *Proc. IEEE VTC2023-Spring*, 2023, doi: 10.1109/VTC2023-Spring57618.2023.10199853.
- [11] 3GPP, "NR; User Equipment (UE) radio transmission and reception; Part 1: Range 1 Standalone," TS 38.101-1 V19.4.0, Release 19, 2026.
- [12] 3GPP, "NR; Physical channels and modulation," TS 38.211 V19.4.0, Release 19, 2026.
- [13] 3GPP, "NR; Multiplexing and channel coding," TS 38.212 V19.4.0, Release 19, 2026.
- [14] 3GPP, "NR; Physical layer procedures for data," TS 38.214 V19.4.0, Release 19, 2026.
- [15] N. Patriciello, S. Lagen, B. Bojovic, and L. Giupponi, "An E2E simulator for 5G NR networks," *Simulation Modelling Practice and Theory*, vol. 96, Art. no. 101933, 2019, doi: 10.1016/j.simpat.2019.101933.
- [16] G. Ferreira, B. Bojovic, A. Larrañaga, A. Ashtari, K. Koutlia, J. Albuquerque, and S. Lagen, "5G-LENA ns-3 nr module," version 5g-lena-v5.0.y, Zenodo, Jul. 2026, doi: 10.5281/zenodo.21165297.